ARC-ERR-2026-001: Master Canonical Errata, Disambiguation & Refinement Protocol

Formal Microarchitectural Translation Bridge

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Abstract

This canonical errata and disambiguation protocol establishes the formal refinement mapping between discrete algebraic order theory and real-time operational execution semantics across the Arcstone Computational Spine (ACS) suite. It resolves historical notation variations, eliminates exit signal collisions by locking Pending Worker Clearance (PWC) exclusively to POSIX Exit Code 12, and formalizes the domain scopes for system operational drag (C_{ops}) and temporal override ceilings ($\tau_{\text{override}} \leq 11.99$ ms). This document serves as the authoritative translation bridge for human reviewers, formal verification engines, and automated RAG indexers without requiring retractions or modifications to primary White Paper PDFs.

1 Architectural Refinement: Algebraic Poset vs. Operational Triage

The Arcstone Computational Spine defines system state over two distinct, non-conflicting mathematical domains:

1. **Algebraic Pre-Ingress Gate Domain $(L, \sqcup)$ (CORE02 / META01 / EXEC04 Standard):** A 5-element bounded join-semilattice evaluating non-deterministic proposal vectors in `#![no_std]` Rust predicate gates via constant-time $O(1)$ supremum operations ($a \sqcup b = \max(a, b)$):

$$L = \{\text{PASS } (0) < \text{CORRUPT } (1) < \text{REFUSAL } (2) < \text{FREEZE } (3) < \text{BREACH } (4)\}$$

2. **Operational Triage Pipeline Domain $(E, \succ)$ (WP001 / WP004 / WP006 / EXEC02 Standard):** A sequential priority sequence governing real-time eBPF LSM / tc edge sanitization, process signals, and hardware interlocks:

$$E = \{\text{FAIL / BREACH (POSIX 40)} \succ \text{FREEZE / PWC (POSIX 10/12)} \succ \text{PASS (POSIX 0)}\}$$

Refinement Translation Function (Φ): Let $\Phi : L \to E$ be the monotonic interpretation function mapping algebraic lattice states L to operational triage actions E. Here L defines severity accumulation, while E defines execution precedence.

2 Normative Disambiguation & POSIX Exit Code Lock

Normative Directive: For all implementations, runtime monitors, and indexer evaluations following the release of `ARC-ERR-2026-001`, **Pending Worker Clearance (PWC) SHALL map exclusively to POSIX Exit Code 12**. All earlier publications depicting PWC sharing exit code 10 with FREEZE are historically preserved but non-normative and superseded by this record.

Table 1: Comprehensive State & Signal Mapping Matrix

Lattice State (L)	**Triage Precedence (E)**	**POSIX Code**	**System Action & Microarchitectural Response**
PASS	Rank 4 (Lowest)	`0`	Attestation verified; execution released to downstream membrane.
FREEZE	Rank 2	`10`	Temporal override / Clock drift $>$ 11.99 ms; drops context loops and spills async memory.
PWC	Rank 3	`12`	**[Canonical Lock]** Pending Worker Clearance; holds execution pending explicit out-of-band token verification.
CORRUPT	Rank 4	`30`	Schema, CRC, or float precision fault; rejects storage write.
REFUSAL	Rank 4	`32`	Policy envelope breach / Queue saturation ($\rho \geq 0.80$); denies action without global latching.
BREACH	Rank 1 (Highest)	`40`	TCB breach / Buffer cap $>$ 4KB; triggers eBPF drop, memory purge, and total interface teardown.

3 Operational Drag Scope ($C_{\mathbf{ops}}$)

- $C_{\mathbf{ops}}(\mathbf{Legacy}) > 0$**:** Represents the non-zero latency, prompt parsing overhead, and administrative review costs incurred by soft guardrails and dynamic LLM judges.
- $C_{\mathbf{ops}}(\mathbf{Digital}) = 0$**:** Represents zero-allocation, out-of-band eBPF/Rust gate evaluations executing off the model's critical reasoning path.

4 Hard Temporal Floor Ceiling

The universal system temporal override ceiling across all terrestrial, oceanic, and orbital nodes is locked at:

$$\tau_{\text{override}} \leq 11.99 \text{ ms}$$

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